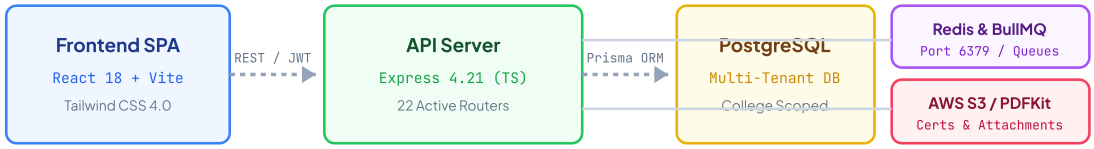
FORGE LOOM LMS

Enterprise Technical Documentation

# Full Platform System Architecture, Wireframes & Operational Guide

A definitive, production-accurate manual detailing the active technology stack, multi-tenant college isolation, role journeys, UI wireframes, seed credentials, and backend API endpoints.




**OFFICIAL REPOSITORY**
`github.com/Winnovationforge/Forge-loom`

**ARCHITECTURE MODEL**
Monorepo (Client, Server, Shared Types)

**RUNTIME VERIFICATION**
API: Port 4000 | Client: Port 5173 | DB: 5433

**DEFAULT SEED PASSWORD**
`test1234` (Universal Across All 11 Roles)

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## 1 Active Technology Stack

The **FORGE Loom LMS** platform is structured as a cohesive, high-performance monorepo with three interconnected packages: `apps/client`, `apps/server`, and `packages/shared-types`. The stack eliminates obsolete dependencies and focuses entirely on production-active modules.

### Frontend Client ( `apps/client` )

Port 5173

Modern single-page application built for responsive, low-latency micro-interactions:

| Component | Implementation & Scope |
|---|---|
| **Core Engine** | `React 18.3.1` with `TypeScript 5.6.3` for strict end-to-end type safety. |
| **Build Tool** | `Vite 5.4.8` with Instant Hot Module Replacement (HMR). |
| **Styling & Design** | `Tailwind CSS v4.0` with custom glassmorphism design tokens, royal blue accent, and responsive flex/grid layouts. |
| **Routing** | `React Router DOM 6.27.0` with dynamic `RequireAuth`, role-guarded nesting ( `RequireRole` ), and `ROLE_HOME_PATH` redirection. |
| **Data Caching** | `@tanstack/react-query 5.59.16` handling server cache, refetching, and optimistic mutations. |
| **Asset Suite** | `Lucide React 0.454.0` icons combined with custom flat 2D vector graphic illustrations ( `.svg` ). |

### Backend API Server ( `apps/server` )

Port 4000

Enterprise REST service layered with strict validation, security headers, and relational isolation:

| Component | Implementation & Scope |
|---|---|
| **Runtime** | `Node.js v22 (ESM)` executed natively using `tsx watch` for development and `tsc` production build. |
| **HTTP Framework** | `Express 4.21.1` with `helmet` security headers, `cors` with credentials, `cookie-parser`, and `morgan` dev logger. |
| **ORM & Database** | `Prisma ORM 5.22.0` querying a multi-tenant `PostgreSQL` database (Port 5433). |
| **Caching & Queues** | `ioredis 5.4.1` connecting to Redis (Port 6379) and `BullMQ 5.21.2` for asynchronous event/email processing. |
| **Authentication** | `jsonwebtoken (JWT) 9.0.2`, `bcryptjs 2.4.3` password hashing, and cookie session transport. |
| **Validation** | `Zod 3.23.8` request body, param, and query schemas. |
| **Media & Certs** | `@aws-sdk/client-s3` + `PDFKit 0.20.1` & `qrcode 1.5.4` for on-the-fly certificate rendering. |

### Shared Monorepo Domain Package ( `packages/shared-types` )

Zero Duplication

Ensures perfect compile-time harmony between frontend UI models and backend Prisma schemas. Defines 11 distinct user roles ( `Role` enum: Student, Mentor, Trainer, Speaker, Hr, Sponsor, CollegeAdmin, CommunityLeader, MediaPartner, Member, ForgeAdmin, CourseAdmin), Course Progress structures, Booking statuses, Sprint models, and Certificate verification payloads.

## 2 Active Platform Modules (Production Features Only)

Per architectural guidelines, only currently implemented, functional modules mounted in `apps/server/src/app.ts` and configured in client routes are listed. Unused legacy stubs have been excluded.

| Active Module | Mounted API Route | Core Capability & Business Logic |
|---|---|---|
| **Authentication & Identity** | `/api/auth` | User login, registration with automatic college provisioning (for College Admin), JWT cookie generation, session verification, and logout. |
| **Course & Curriculum** | `/api/courses` `/api/catalog` | Course creation/editing by Course Admins, public catalog browsing, module/topic syllabus hierarchies, and prerequisite checks. |
| **Enrollments & Billing** | `/api/enrollments` | Batch enrollments, student course registration, college cohort allocation, and optional Razorpay integration client hooks. |
| **Video & Course Progress** | `/api/video-progress` `/api/students` | Timestamp-accurate YouTube/video playback tracking, completed lesson flags, student progress rollups, and weekly learning velocity tracker. |
| **Sessions & Attendance** | `/api/sessions` `/api/students/attendance` | Live trainer/mentor lectures, batch session scheduling, student attendance check-ins, and aggregate attendance percentage calculations. |
| **Citadel Incubator & Sprints** | `/api/sprints` `/api/problem-statements` `/api/teams` | Core innovation engine: 3-stage sprint lifecycles (Sprint 1, 2, 3), student team grouping, industry challenge assignment, milestone submissions, and mentor reviews. |
| **Mentor Booking Engine** | `/api/bookings` | 1-on-1 expert advisory bookings, available time slot querying, glassmorphic scheduling modal, session confirmation, and feedback logging. |
| **Certificates & Verification** | `/api/certificates` | Automated dynamic PDF certificate generation with embedded cryptographic verification QR codes and public lookup endpoint ( `/verify/:token` ). |
| **Events & Workshops** | `/api/events` | Hackathons, masterclasses, and guest lectures. Supports attendee registration, event capacity constraints, and community RSVP. |
| **HR Talent Pool Pipeline** | `/api/talent-pool` `/api/hr-profile` | Corporate recruiter dashboard, company profile builder, student skill filtering, portfolio explorer, and interview shortlisting. |
| **Multi-Tenant College Admin** | `/api/colleges` `/api/cohorts` | Institution-level tenant isolation: student rosters, faculty/trainer tracking, department benchmarks, and cohort management. |
| **Forge Global Admin** | `/api/admin` | Super-admin controls: platform-wide KPIs, college onboarding, cross-institutional user management, and system health metrics. |
| **Real-Time Notifications** | `/api/notifications` | User notification feed, unread bell count badge, milestone status updates, and session booking alerts. |

# 3 Complete LMS Wireframe & Layout Architecture

The LMS follows a unified **Global Shell Architecture** across all 11 user roles. This ensures visual consistency while dynamically swapping the sidebar navigation and content region based on authenticated role permissions.

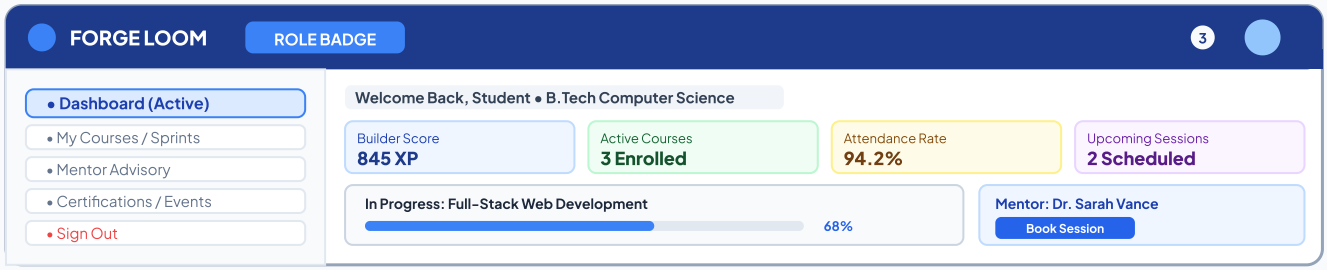



FIGURE 3.1: GLOBAL APPLICATION SHELL & STUDENT DASHBOARD WIREFRAME ARCHITECTURE

### Student Portal ( `/student/*` )

Core Learner

- **Home Dashboard ( `/student` ):** Personal greeting, Builder Score counter, 4 KPI cards, active course progress bars, upcoming mentor sessions, and next events.
- **Course Player ( `/student/courses/:id` ):** Interactive syllabus navigation tree, timestamp-synced video player, lesson completion checkbox, and downloadable resources.
- **Citadel Hub ( `/student/citadel` ):** Innovation incubator: 3-Sprint horizontal progress stepper, Problem Statement challenge viewer, milestone submission forms, and mentor review threads.
- **Book a Mentor ( `/student/mentor` ):** Glassmorphic mentor profile modal, date/time slot picker, topic definition, and booking confirmation.

### Mentor Portal ( `/mentor/*` )

Advisory & Evaluation

- **Mentor Dashboard ( `/mentor` ):** Pending 1-on-1 booking requests, assigned mentee rosters, upcoming sessions calendar, and sprint review queues.
- **Assigned Mentees ( `/mentor/students` ):** Drill-in student profiles, course completion rates, attendance history, and direct notes.
- **Citadel Teams ( `/mentor/teams` ):** Innovation teams milestone grading, feedback submission box, and investor readiness approvals.
- **Availability Manager ( `/mentor/sessions` ):** Weekly calendar slot configuration, accept/reject booking requests.

### HR & Talent Portal ( `/hr/*` )

Recruitment Pipeline

- **Recruiter Home ( `/hr` ):** Active job opportunities, shortlisting stats, campus hiring event attendance.
- **Talent Pool ( `/hr/talent-pool` ):** Full-text student candidate search, multi-skill selection tags, Builder Score range slider, and portfolio links.
- **Builder Profile Drill-down:** Detailed verified resume: completed projects, GitHub links, sprint evaluations, and interview scheduling action.

### College & Forge Admin ( `/college` & `/admin` )

Institutional Control

- **College Admin ( `/college` ):** Institution-isolated dashboard (Forge Institute vs Riverside College), student enrollment counts, faculty rosters, and placement metrics.
- **Course Admin ( `/course-admin` ):** Rich course editor, curriculum builder, lesson module creator, and batch assignments.
- **Forge Super Admin ( `/admin` ):** Cross-college global analytics, new college onboarding, system health, and database metrics.

## 4 Seed Passwords, User Directory & Multi-Tenant Clusters

The platform includes automated deterministic database fixtures seeded via `apps/server/src/scripts/seed.ts`. All seed accounts share a single, universal development password.

**UNIVERSAL SEED CREDENTIALS**

**Password:** `test1234`

Applies to all 11 user roles across both multi-tenant college clusters. Bcrypt hashed & salted at seed time.

### 4.1 Multi-Tenant College Clusters (Tenant Isolation Testing)

Two isolated colleges are seeded in PostgreSQL. Users in **Cluster 1** (Forge Institute) are strictly isolated from users in **Cluster 2** (Riverside College), verifying the multi-tenant security barrier.

| College / Tenant | Role | Display Name | Login Email | Landing Route |
|---|---|---|---|---|
| **College Cluster 1** Forge Institute of Technology | CollegeAdmin | Forge Institute of Technology | `college1@forgeloom.dev` | `/college` |
| | Student | Student One | `student1@forgeloom.dev` | `/student` |
| | Mentor | Mentor One | `mentor1@forgeloom.dev` | `/mentor` |
| | Trainer | Trainer One | `trainer1@forgeloom.dev` | `/trainer` |
| **College Cluster 2** Riverside College of Engineering | CollegeAdmin | Riverside College of Engineering | `college2@forgeloom.dev` | `/college` |
| | Student | Student Two | `student2@forgeloom.dev` | `/student` |
| | Mentor | Mentor Two | `mentor2@forgeloom.dev` | `/mentor` |
| | Trainer | Trainer Two | `trainer2@forgeloom.dev` | `/trainer` |

### 4.2 Global & Cross-Ecosystem Accounts

Platform-wide roles governing curriculum authoring, corporate placement, industry keynotes, and administrative governance:

| Role | Designation / Entity | Login Email | Universal Password | Landing Route |
|---|---|---|---|---|
| ForgeAdmin | Forge Platform Super Admin | `admin1@forgeloom.dev` | `test1234` | `/admin` |
| CourseAdmin | Lead Curriculum Creator | `course_admin1@forgeloom.dev` | `test1234` | `/course-admin` |
| Hr | Corporate Hiring Partner | `hr1@forgeloom.dev` | `test1234` | `/hr` |
| Speaker | Guest Industry Keynote | `speaker1@forgeloom.dev` | `test1234` | `/speaker` |
| Sponsor | Corporate Innovation Sponsor | `sponsor1@forgeloom.dev` | `test1234` | `/sponsor` |
| CommunityLeader | Tech Chapter Organizer | `community1@forgeloom.dev` | `test1234` | `/community` |
| MediaPartner | Media Coverage Representative | `media1@forgeloom.dev` | `test1234` | `/media` |

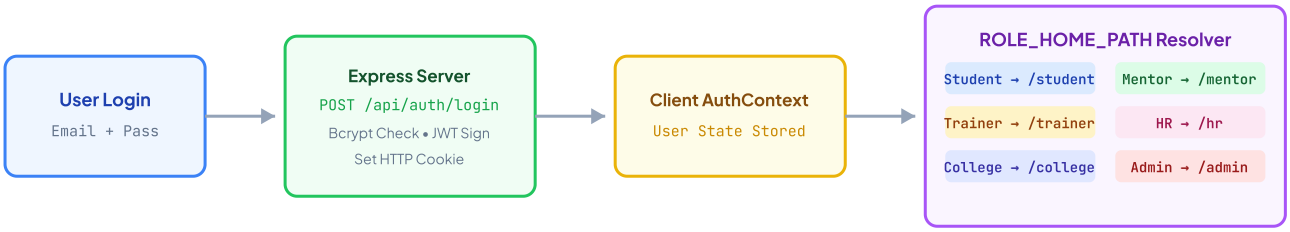
| Role | Designation / Entity | Login Email | Universal Password | Landing Route |
|---|---|---|---|---|
| Member | General Ecosystem Member | `member1@forgeloom.dev` | `test1234` | `/member` |

# 5 Operational Flow Diagrams & User Journeys

The following diagrams illustrate the core lifecycle operations of FORGE Loom LMS, from role authentication and dynamic landing resolution to the Citadel innovation sprint reviews and mentor scheduling.

### Diagram 5.1: Authentication & Dynamic Role-Based Routing Flow

Security Layer




On successful login, the frontend receives the user's role and immediately redirects them to their specialized portal via `roleHome.ts`. Direct deep-link navigation is guarded by `RequireRole` to prevent unauthorized privilege escalation.

### Diagram 5.2: Student Citadel Innovation Sprint & Mentorship Journey

Learning Lifecycle




Progressing through course modules increases the student's Builder Score. In parallel, team-based Citadel Sprints test practical problem-solving. Completing milestones unlocks mentor review feedback, automated certificate issuance, and exposure to corporate recruiters in the HR Talent Pool.

### Diagram 5.3: 1-on-1 Mentor Booking Lifecycle

Advisory Pipeline

1. **Slot Discovery:** Student opens the Mentor Booking Modal on `/student/mentor`. Frontend queries available slots for assigned mentors via `GET /api/bookings/available-slots`.
2. **Booking Request:** Student specifies meeting agenda, topic, and preferred time slot. Submits `POST /api/bookings`, which creates a pending booking record.
3. **Notification Trigger:** Backend issues an in-app notification via `/api/notifications` alerting the mentor to the new booking request.
4. **Mentor Confirmation:** Mentor logs into `/mentor/sessions`, reviews the student's current Builder Score and project status, and approves or reschedules the session.
5. **Session Execution & Feedback:** Post-session, the mentor submits session notes and evaluation metrics, updating the student's advisory log.

## 6 System Operations, Verification & Maintenance

Essential command-line operations for starting services, re-seeding fixtures, executing database migrations, and validating compile-time TypeScript integrity across the entire monorepo.

### Development Startup & Service Commands

Local Environment

| Task | Command | Description |
|---|---|---|
| **Full Stack Dev Run** | `npm run dev` | Builds `@forge-loom/shared-types` and executes both `apps/client` (Vite: 5173) and `apps/server` (Express: 4000) concurrently. |
| **Seed Database** | `npm run seed --workspace=@forge-loom/server` | Seeds all 11 user roles across Forge Institute and Riverside College with password `test1234`. Safe to re-run (idempotent). |
| **Typecheck Entire Repo** | `npm run typecheck` | Executes `tsc --noEmit` across client, server, and shared types ensuring zero type errors. |
| **Database Migrations** | `npx prisma migrate dev --schema=apps/server/src/prisma/schema.prisma` | Applies schema updates to PostgreSQL and regenerates the Prisma Client. |
| **Docker Services** | `docker compose -f infra/docker-compose.yml up -d` | Launches PostgreSQL (port 5433) and Redis (port 6379) background containers. |

### Summary & Repository Verification

This document represents the complete, verified state of the **FORGE Loom LMS** platform. The entire codebase is stored and tracked under the official repository:

`https://github.com/Winnovationforge/Forge-loom.git`

All visual assets conform to the 2D flat modern aesthetic, free of emojis, and optimized for scalable institutional deployment across colleges, universities, and enterprise skill centers.